

Quantifying the Role of External Reflection in Laser Mode Hopping for Heat-Assisted Magnetic Recording

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Heat-assisted magnetic recording (HAMR) employs a laser diode integrated with an optical waveguide and a plasmonic antenna, namely near field transducer (NFT), in the recording head to heat to localize and enhance heating of the recording media. External optical feedback from interfaces within the slider can perturb the laser's threshold gain, thereby affecting mode-hop behavior and optical power stability. In this work, we model the HAMR optical path as a three-mirror cavity system and quantify the influence of external reflectance and cavity length on longitudinal mode hopping. Our quantitative analysis shows that although the feedback level in HAMR remains in the weak regime (feedback coefficient (C) < 1), it modulates the threshold gain into a wavelength-dependent function, significantly altering the mode-hop pattern and amplitude. We demonstrate that mode-hop amplitude is significantly reduced by lowering the reflectance from the recording head back to the laser. Additionally, the mode-hop behavior is tuned by adjusting the waveguide length inside the recording head: shorter waveguide lengths reduce the frequency of large wavelength hops for a given temperature variation.

Index Terms— heat-assisted magnetic recording, mode hopping, reflection, threshold gain

I. INTRODUCTION

Heat-assisted magnetic recording (HAMR) utilizes laser light to locally heat the recording media, enabling writing on high-coercivity granular media. A high-fidelity HAMR recording system requires laser diode to provide stable and reliable optical power during data writing process. Achieving a stable laser power during HAMR operation is extremely challenging and has been a key gating item in achieving high performance for HAMR HDDs. The laser power variation is mostly attributed to the reflection-driven instability and the temperature change of the laser diode. There has been significant effort minimizing laser temperature change during turn on to eliminate mode hop [1]–[3]. Here, the focus is to investigate the role of feedback or external reflection. Laser diode response is sensitive to feedback from the components integrated inside the slider. The severity of feedback-induced instability depends on both the reflectance and the external cavity length; for a given reflectance, a longer external cavity produces stronger feedback effects [4]–[7]. These effects have been extensively studied for telecom applications as well where the reflections originate from far more distant reflectors. In a HAMR recording head, the external cavity length is limited by the slider geometry to approximately 150–200 μm , which is much shorter than the typical fiber length in telecom applications (0.5–2 m). For this reason, feedback effects have received comparatively little attention in the HAMR context.

In this work, we investigate the impact of external optical reflections on mode-hop behavior in HAMR recording heads through a combination of modeling and experimental validation. The results demonstrate that external reflections play a critical role in governing mode-hop dynamics. Reducing the

reflectance from the recording head back into the laser cavity significantly suppresses the amplitude of mode hops.

Furthermore, we show that mode-hop behavior can be systematically engineered by modifying the internal waveguide length of the recording head. Specifically, shorter waveguide lengths reduce the frequency of large wavelength hops under a given temperature variation. These findings highlight viable design strategies for improving wavelength stability and overall performance in HAMR systems.

II. THREE-MIRROR CAVITY MODEL

We model the HAMR optical path as a three-mirror system, illustrated in Fig. 1. Here, r_1 and r_2 denote the amplitude reflection coefficients of the back and front facets of the laser diode, respectively, and r_3 denotes the reflection coefficient at the external reflective interface. In the absence of external feedback, the front-facet reflectivity is determined solely by r_2 . When feedback is present, light reflected from the external cavity re-enters the laser, and the combined effect of r_2 and r_3 can be described by an effective reflection coefficient r_2' given by (1) which is the first-order approximation assuming the higher-order reflection is much weaker [8].

$$r_2'(\lambda) = r_2 + (1 - |r_2|^2)r_3 \exp(i \frac{2\pi}{\lambda} 2nL_{ext}) \quad (1)$$

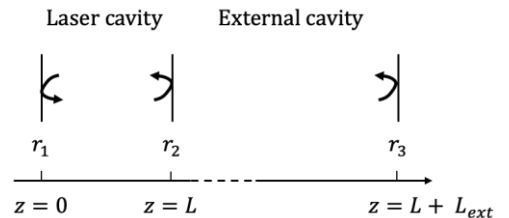


Fig. 1. Three-mirror cavity model for a laser diode with external feedback. r_1 and r_2 are the amplitude reflection coefficients at the back facet ($z=0$) and front

facet ($z = L$) of the laser cavity, respectively, and r_3 is the reflection coefficient at the external interface ($z = L + L_{\text{ext}}$).

At threshold, the round-trip gain equals the round-trip loss. Replacing r_2 with r_2' modifies the threshold gain condition, as described by (2), where α_s is the loss inside the laser cavity including absorption and scattering. The change in threshold gain due to external feedback is then given by (3).

$$g_{th} = \alpha_s + L^{-1} \ln(1/|r_1||r_2'|) \quad (2)$$

$$\Delta g_{th} = L^{-1} \ln(|r_2|/|r_2'|) \quad (3)$$

TABLE I
PARAMETERS USED IN THREE-MIRROR CAVITY MODEL

Parameter	Description
α	Linewidth enhancement factor
n	External cavity refractive index
$L_{\text{ext}} (\mu\text{m})$	External cavity length

III. RESULTS AND DISCUSSION

A. Threshold Gain and Phase Modulation

Fig. 2(a) shows the wavelength-dependent threshold gain in the presence of external feedback. Because light travels an additional path through the external cavity, the round-trip phase is modified. The change in phase relative to the case without feedback is given by (4).

$$\Delta\phi = -\alpha \ln\left(|r_2|/|r_2'|\right) + \left(2\pi/\lambda - 2\pi/\lambda_0\right) 2nL_{\text{ext}} + \arg(r_2') \quad (4)$$

Fig. 2(b) presents the $\Delta(\text{phase})$ – $\Delta(\text{gain})$ diagram using the parameters in Table I, which approximate those of a typical HAMR laser diode. The curve is nearly cosinusoidal, and the feedback coefficient C [7]–[8] is less than unity. This confirms that the system operates in the weak-feedback regime [7], well below the onset of coherence collapse. In other words, mode hopping in this regime is not intrinsically driven by reflection-induced chaos but rather by the thermal drift of the gain peak due to laser self-heating. As r_3 increases, the system eventually enters the chaotic regime [7], where the dominant lasing mode hops unpredictably among longitudinal modes. A key diagnostic for distinguishing feedback-induced mode hopping from thermally driven hopping is whether the dominant mode stabilizes once the laser reaches thermal equilibrium.

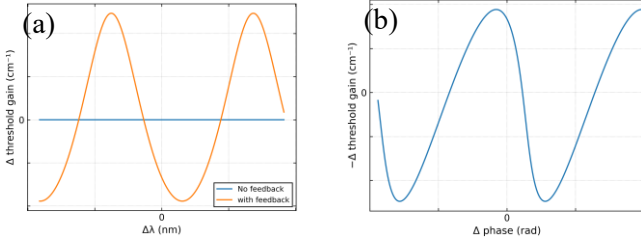


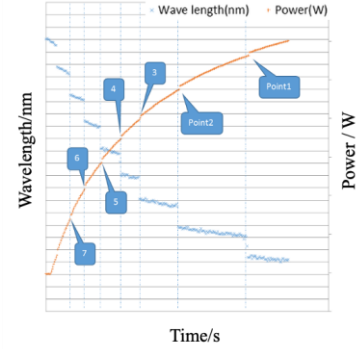
Fig. 2. (a) Change in threshold gain as a function of wavelength detuning with and without external feedback. Without feedback, the threshold gain is wavelength-independent; with feedback, it is modulated approximately as a cosine function. (b) Negative change in threshold gain versus round-trip phase change, showing near-cosinusoidal behavior characteristic of the weak-feedback regime ($C < 1$).

B. Effect of External Reflection on Mode-Hop Pattern

Having established that HAMR operates in the weak-feedback regime, we note that thermal self-heating nevertheless causes mode hopping. As the junction temperature rises, the gain peak red-shifts due to bandgap narrowing in the III–V quantum wells. The longitudinal mode best aligned with the gain peak experiences the highest amplification and becomes dominant; successive modes take over as the temperature continues to rise.

Without external reflection (laser diode only), the resulting mode-hop pattern is almost uniform in both wavelength and optical power, and the hop amplitude in power is very small, as shown in Fig. 3(a). With external reflection, however, the hop pattern is altered, and the power-hop amplitude increases substantially because the feedback modulates the threshold gain, as shown in Fig. 3(b).

(a) Mode hop w/o external feedback



(b) Mode hop with external feedback

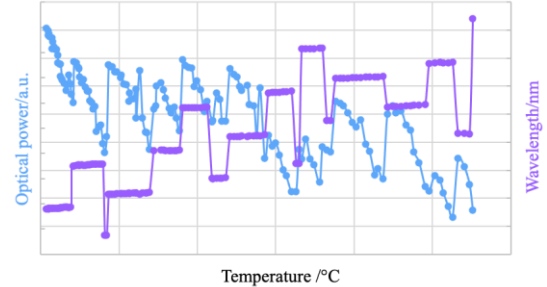


Fig. 3. (a) Measured wavelength (blue) and optical power (orange) as a function of time during laser self-heating without external feedback. The mode hops are uniform in wavelength spacing and produce small power variations. Numbered labels indicate successive longitudinal mode transitions. (b) Measured optical power and wavelength as a function of temperature with external feedback. The mode-hop pattern becomes irregular, with significantly larger power-hop amplitudes due to feedback-induced modulation of the threshold gain.

C. Influence of Reflectance and Cavity Length

Fig. 4(a) compares mode-hop patterns under different reflection conditions at the slider interface. The blue curve corresponds to the original (uncoated) interface with higher reflectance, and the purple curve corresponds to an interface

with an antireflection coating and therefore lower reflectance. A clear reduction in hop amplitude is observed for the coated interface, consistent with the reduced modulation depth of the threshold gain.

Fig. 4(b) compares mode-hop patterns for a short external cavity (waveguide length inside the slider) and a longer external cavity (waveguide length inside the slider). For the same temperature rise, large wavelength hops occur less frequently with a shorter external cavity because the period of the threshold gain modulation is inversely proportional to the external cavity length. Fig. 4(c) shows the variation of threshold current with temperature. The threshold current exhibits a cosine-like modulation superimposed on an increasing trend, accompanied by discrete hops. Because the hops disrupt the periodicity, the apparent period extracted from the threshold current data does not correspond to the true period of the underlying cosine modulation of the threshold gain.

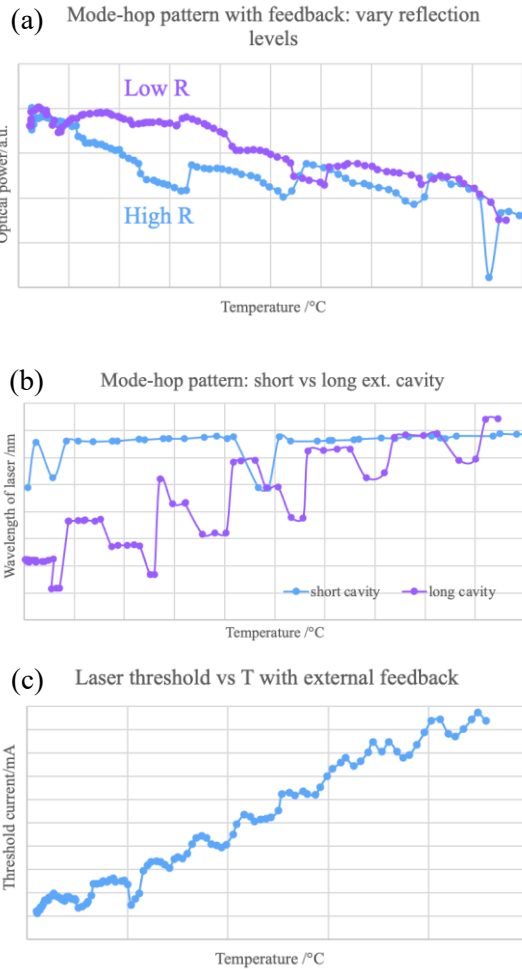


Fig. 4. (a) Optical power as a function of temperature for a high-reflectance interface (blue) and a low-reflectance interface (purple). (b) Lasing wavelength as a function of temperature for a short external cavity (blue) and a long external cavity (purple). (c) Threshold current as a function of temperature in the presence of external feedback.

IV. CONCLUSION

External optical feedback, even in the weak regime, significantly affects mode-hop behavior in HAMR laser diodes. Although the coupling factor C remains below unity—below

the threshold for chaotic instability—the feedback modulates the threshold gain from a constant value to a wavelength-dependent cosinusoidal function. This modulation distorts the otherwise uniform mode-hop pattern and increases the power-hop amplitude. Reducing the reflectance at the waveguide interface correspondingly reduces the hop amplitude, and a shorter external cavity length decreases the frequency of large wavelength hops over a given temperature range. These findings provide design guidelines for minimizing reflection-induced power instability in HAMR systems.

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